

WELL COMPLETION REPORT (SYNTHETIC)

OIL-HPJ-005 - Hapjan-005

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000055
Field / District	Hapjan / Tinsukia
Block	OALP-III-07
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	1999-08-16 / 2000-01-08
Measured / TVD	3125 m / 3080 m
Trajectory	Vertical; inc 4.1 deg; az 37.2 deg
Formation Interval	1477-2635 m
Lithology	Fine sandstone with carbonaceous shale
Pore Pressure Reference	3495 psi
Fracture Gradient	15.15 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (53.8/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	781	Surface
9 5/8 in casing	1812	Intermediate
7 in liner/casing	2750	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
1999-10-19 1525 m	Mud Loss	Critical	Flow-out dropped below flow-in with pit loss. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
1999-12-06 2481 m	Mud Loss	Critical	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
1999-09-19 882 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Check solids loading and BHA hydraulics before drilling ahead. Lesson: Correlate pressure anomalies with flow before escalating risk.
1999-10-22 1577 m	Kick	High	Pit gain and flow with pumps off were observed. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
1999-10-31 1761 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
1999-11-27 2303 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.
1999-12-14 2638 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.

Completion Summary

The synthetic well reached 3125 m MD in the Barail interval. The NWIS knowledge record links 19 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.